[14X030] Introduction to Computational Finance

Exercise # 7

April 14, 2026

General instructions

Each student is expected to upload on Moodle a **zip** file named as *firstname_lastname*, containing the followings:

- A report in **pdf** format to answer exercise questions.
- The **code** used to generate the results of the report.
- If preferred, submit a unique **Jupyter notebook** with both code and report explanations.

Deeper instructions

- **Report**:
  - Answers to the TP questions. Figures and numerical results are necessary but **strictly not** sufficient: provide your analysis/comments as well.
  - Example: if your results surprise you, explain why and explain what the expected result was.
  - **Pay attention** to the presentation: axis labels, legend, etc.
- **Code**:
  - Implementation questions + code used to produce the results and figures in the report.
  - Programming language of your choice.
  - Do not use ChatGPT, or any other LLM/AI-assisted tools while writing down your code. Try by yourself, otherwise it will be counted as failed.
  - I **strongly recommend Python/Julia**: I won't be able to assist you in the same way if you use a different language.

Deadline

Any questions to: Lorenzo Bini.
Upload on Moodle due by : **April 20, 2026 at 11:59 pm**

Optimal Portfolio

In this series we look at the closing prices of Macdonald's, Bank of America, IBM, Chevron, Coca-Cola, Novartis and AT&T, over a one-year time span extending from 2013-05-01 to 2014-05-01.

Download the file `closing_prices.csv` on Moodle. You can load it with the following Python code:

```
import pandas as pd
df = pd.read_csv('closing_prices.csv')
data = df.to_numpy() # if you prefer working with np array
```

We define the weight vector of the portfolio $w = \{w_1, ..., w_7\}$ such that $\sum_{i=1}^{7} w_i = 1$.

1. Write a function that estimates the expected return and the risk (standard deviation of returns) for a given weight vector. Then, plot the return against the risk for 100000 randomly chosen weights vectors (Monte-Carlo simulation). What do you observe?

2. We introduced the following analytical expressions during the course to compute the weight vector that minimizes the risk given a desired portfolio return μ_p:

$$\begin{aligned} a &= \mathbf{1}^T C^{-1} \mathbf{1} \\ b &= \mathbf{1}^T C^{-1} \mu \\ c &= \mu^T C^{-1} \mu \\ d &= ac - b^2 \\ \lambda_1 &= \frac{c - b\mu_p}{d} \\ \lambda_2 &= \frac{a\mu_p - b}{d} \\ w &= C^{-1}(\lambda_1 \mathbf{1} + \lambda_2 \mu) \end{aligned} \tag{1}$$

   With $\mu = \{\mu_1, ..., \mu_7\}$ the expected returns of each stock and C the covariance matrix of returns.
   Using this expression, draw Markowitz's efficient frontier for portfolio return $\mu_p \in [-0.0006, +0.0004]$.

3. Using the efficient frontier, find the weight of the portfolio with the minimal volatility. What can you say about the return of this portfolio?

Optimal Portfolio under No-Short Selling Constraints

At this point, you will extend the optimal portfolio analysis by imposing a no-short selling constraint (i.e. all portfolio weights must be non-negative). You will compute the efficient frontier under this additional constraint and compare it to the unconstrained (Markowitz) efficient frontier.
Keep using the provided `closing_prices.csv` file which contains the closing prices of Mcdonald's, Bank of America, IBM, Chevron, Coca-Cola, Novartis, and AT&T over one year (from 2013-05-01 to 2014-05-01). Load the data and compute the daily returns for each stock.

1. Formulate the portfolio optimization as a minimization problem where the objective is to minimize the portfolio variance subject to the following constraints:
    - The portfolio return is equal to a given target μ_p.
    - The sum of weights is equal to 1.
    - All weights are non-negative (no short selling).
2. Use a suitable Python optimization library (e.g., `cvxpy`) to solve the constrained problem.
3. Plot the constrained efficient frontier over the same range of previous μ_p.
4. Compare the unconstrained (computed from Exercise # 1) and constrained efficient frontiers.
5. Discuss the impact of the no-short selling constraint on the risk-return trade-off.
6. Identify and report the portfolio with minimal risk under the no-short selling constraint and comment on its expected return.

Note: You can use the `cvxpy` library to solve the constrained optimization problem. The library allows you to define the optimization variables, objective function, and constraints in a straightforward manner. Make sure to install it if you haven't already by `pip install cvxpy`.